

Drivers of China's high-quality development: The role of intangible factors[☆]

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ABSTRACT

China's economy is shifting from high-growth to high-quality development, the core of which is promoting total factor productivity (TFP), especially with intangible factors. However, the role of intangible factors has been neglected in the literature. We investigate the drivers of TFP and identify the contribution of intangible factors in China by constructing a global nonradial Luenberger productivity indicator (GLPI) and incorporating intangible factors (knowledge, data, and technology). The intangible factors are constructed through factor analysis using 37 indicators. We find that technological innovation is the largest driver of TFP change. Second, the contribution of intangible factors is larger than that of traditional factors (labor, capital, and energy). Intangible factors such as knowledge, data, and technology increase TFP by about 4.2%, 0.32%, and 5%, respectively, while contributing to reducing carbon emissions. Third, the contribution of intangible factors in TFP growth differs across geographical locations, economic development, and factor richness.

1. Introduction

China's rapid growth in the past few decades has been driven by massive inputs of production factors. However, great changes have taken place in the country's economy and society. At present, China's economy lacks efficiency while heavily polluting the environment. Due to the constantly rising costs of production factors, relying on traditional factors to drive growth is no longer sustainable. Faced with this reality, China is actively promoting an economic shift from the high-growth stage to the high-quality development stage. The report of the 19th National Congress of the Communist Party of China argued that China's economy developed from a high-growth to a high-quality development stage and emphasized improving total factor productivity (TFP), which is a representative indicator of high-quality economic development (Tsionas and Mallick, 2019; Li and Liu, 2015). In recent years, by incorporating environmental pollution and energy input into nonparametric models, a green TFP indicator may be constructed to accurately measure high-quality economic development. On this basis, the Chinese government has proposed promoting high-quality economic development through intangible production factors (such as data, information, technology, and knowledge). A TFP that incorporates these intangible and environmental factors may be a more robust measure of the level of

high-quality economic development than green TFP alone; however, researchers have neglected the contribution of the intangible production factors. Given this, we innovatively construct a new TFP model incorporating the intangible factors and clarify their contributions to TFP change, which is a meaningful and critical issue in achieving high-quality economic development.

Factors of production are constantly changing with the development of the economy and society. William Petty (1662) first proposed the dualism of land and labor as two important factors. As society developed, Say (1803) added capital as a factor, forming the theory of the three factors of production. In October 2019, the Central Committee of the Communist Party of China suggested that China should improve the mechanisms by which the contributions of labor, capital, land, knowledge, technology, management, data, and other factors of production are evaluated by the market and determine remuneration based on their contribution. This means that factors of production have a new complement based on traditional factors of production, such as labor and capital.

These intangible and traditional factors of production are different and have many unique advantages. First, these intangible factors tend to be non-competitive, which mitigates their scarcity to some extent. Second, new similar factors are generated based on the original intangible

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factors, which means that the marginal cost may not increase in the process of constantly inputting intangible factors; that is, the output is increased without any cost of inputs. Third, the input of intangible factors can promote technological innovation to a certain extent, which is an important channel to promote TFP. Fourth, as various intangible factors have just been put into production, they are still in the stage of increasing returns to scale. Fifth, the intangible factors of production tend to cause low pollution, which means that the input of these intangible factors will not have as much negative impact on the environment as traditional factors. Finally, intangible factors can replace traditional factors to a certain extent, which means that output is increased on the premise of reducing costs and negative externalities and leading to the improvement of TFP. Based on the aforementioned reasons, the intangible factors of production have great potential to increase TFP.

Current research on high-quality development can be divided into two categories: theoretical discussions (Pang et al., 2019; Gao et al., 2020) and empirical research, which largely involves the analysis of economic and social variables such as governance (Chen and Chen, 2018) and digital economy (Zhao et al., 2020). To our best knowledge, no study has explored high-quality development from the perspective of production factors while incorporating data and other intangible factors of production.

Several studies explore the contribution of different factors of production. Vinod and Kaushik (2007) found a statistically significant impact of human capital on economic growth in large developing countries across 1982–2001. Based on China's data in the last century, Zhang (1991) used the Cobb–Douglas production function and found that the main driving force of China's economic growth at that time was capital, while the contribution of labor was relatively small. Zhang (2002) believes that developed countries benefit from the world economy and international division of production owing to the differences in the international distribution of elements of the knowledge economy. Bournakis et al. (2018) found that international knowledge spillover plays an important role in promoting output per worker. Müller et al. (2018) used firm-level data to analyze the effects of big data and analytics on firm productivity, finding that big data and analytics can improve the productivity of a firm about 3%–7%.

The research most similar to ours is that of Li and Liu (2015), who studied the spatial-temporal regularities, factor sources, and dynamic evolution of China's regional economic growth performance from 1985

information from 37 Chinese provincial indicators through factor analysis to obtain the factor score of the intangible factors of production (data, knowledge, and technology), avoiding bias caused by measuring intangible factors using a single indicator. Additionally, this avoids the impact of model setting errors on the results, an issue with the parametric method. Second, no studies have systematically and synthetically explored the contribution of intangible and traditional factors of production to Chinese high-quality economic development. Based on the constructed GLPI, we explore the drivers of economic growth in China and obtain the contribution of each intangible factor and technology and efficiency change.

The remainder of this study is organized as follows. Section 2 presents the main research methods. Section 3 describes the data and the measure of variables. Section 4 analyzes the results. The final section is the conclusion and policy implications.

2. Methodology

Considering the complex relationship between the intangible and traditional factors of production, the lack of consensus on the mechanisms by which these intangible factors influence output, and the deviation due to incorrect production function settings, we use the nonparametric method to explore their contributions to TFP, for which no specific production function form is required (Bournakis and Mallick, 2018).¹

We construct the nonparametric model following Zhang (2022), who constructs a biennial nonradial Luenberger productivity indicator (BLPI). However, because the BLPI is not transitive (Pastor et al., 2011), we construct a GLPI that simultaneously incorporates intangible and traditional factors of production and environmental factors to explore their contribution to high-quality economic development. This innovative indicator overcomes the non-transitivity of biennial production technology and the infeasible solution issue of the Luenberger index.

We treat individual provinces in China as the decision-making units, and set the production technology as

$$T = \{(K, L, E, Z, D, T, Y, B) : (K, L, E, Z, D, T) \text{ can produce } (Y, B)\} \# \quad (1)$$

where K is capital, L is labor, E is energy, Z is knowledge, D is data, T is technology, Y is desirable output, and B is undesirable output.

The above formula can be more specifically expressed as

$$T = \left\{ (K, L, E, Z, D, T, Y, B) : \sum_{n=1}^n h_n K_n \leq K, \sum_{n=1}^n h_n L_n \leq L, \sum_{n=1}^n h_n E_n \leq E, \sum_{n=1}^n h_n Z_n \leq Z, \right. \\ \left. \sum_{n=1}^n h_n D_n \leq D, \sum_{n=1}^n h_n T_n \leq T, \sum_{n=1}^n h_n Y_n \geq Y, \sum_{n=1}^n h_n B_n = B, h_n \geq 0, n = 1, \dots, n \right\} \# \quad (2)$$

to 2012 using a nonparametric model. However, we use global production technology, which better avoids the problem of infeasible solutions, and the results are more comparable. Furthermore, this study covers 2010–2019, when China's economy was facing transformation and upgrading, namely, a period of great changes. It is thus necessary to continue to analyze the issue of the driving forces of TFP growth in China. Moreover, Li and Liu (2015) did not analyze the contribution of the intangible production factors, while this study incorporates these factors into the traditional nonparametric model to analyze the contribution of intangible factors of production to TFP growth.

This study makes two major contributions. First, we construct a global nonradial Luenberger productivity indicator (GLPI) that simultaneously incorporates intangible factors of production and energy and environmental factors. For the intangible factors, we extracted

where n means n provinces and h_n is an intensity variable to indicate variable or constant returns to scale.

Subsequently, we calculate the global nonradial directional distance function using the following model,

$$\bar{D}_N^{Gt}(K^t, L^t, E^t, Z^t, D^t, T^t, Y^t, B^t; g^t) = \text{Max} \left[\beta_N^{Gt} : \beta_N^{Gt} = w_l \beta_{NL}^{Gt} + w_k \beta_{NK}^{Gt} \right. \\ \left. + w_e \beta_{NE}^{Gt} + w_z \beta_{NZ}^{Gt} + w_d \beta_{ND}^{Gt} + w_t \beta_{NT}^{Gt} + w_y \beta_{NY}^{Gt} + w_b \beta_{NB}^{Gt} \right]$$

¹ In addition, we also set a parametric model to explore the effects of intangible factors on China's economic high-quality development; the results are consistent with the nonparametric model.

$$s.t. \sum_{n=1}^n h_n^t K_n^t \leq K_N^t - \beta_{NK}^{Gt} g_{NK}^t,$$

$$\sum_{n=1}^n h_n^t L_n^t \leq L_N^t - \beta_{NL}^{Gt} g_{NL}^t,$$

$$\sum_{n=1}^n h_n^t E_n^t \leq E_N^t - \beta_{NE}^{Gt} g_{NE}^t,$$

$$GLPI = GLPI_K + GLPI_L + GLPI_E + GLPI_Z + GLPI_D + GLPI_T + GLPI_Y + GLPI_{B\#}(7)$$

$$\sum_{n=1}^n h_n^t Z_n^t \leq Z_N^t - \beta_{NZ}^{Gt} g_{NZ}^t,$$

$$\sum_{n=1}^n h_n^t D_n^t \leq D_N^t - \beta_{ND}^{Gt} g_{ND}^t,$$

$$\sum_{n=1}^n h_n^t T_n^t \leq T_N^t - \beta_{NT}^{Gt} g_{NT}^t,$$

$$\sum_{n=1}^n h_n^t Y_n^t \geq Y_N^t + \beta_{NY}^{Gt} g_{NY}^t,$$

$$\sum_{n=1}^n h_n^t B_n^t \leq B_N^t - \beta_{NB}^{Gt} g_{NB}^t \quad \# \quad (3)$$

where the superscript t is time. Furthermore, w is the weight of each variable, which is summed to 1, and following Zhou et al. (2012), we set the weight of the input, desirable output, and undesirable output to 1/3 each. Thus, $w = (w_k, w_l, w_e, w_z, w_d, w_t, w_y, w_b) = (1/18, 1/18, 1/18, 1/18, 1/18, 1/18, 1/3, 1/3)$ in this study.

Based on the results of equation (3), following Oh (2010) and Chambers et al. (1996), we construct the GLPI as follows:

$$GLPI_n = \bar{D}_n^G(K^t, L^t, E^t, Z^t, D^t, T^t, Y^t, B^t; g) - \bar{D}_n^G(K^{t+1}, L^{t+1}, E^{t+1}, Z^{t+1}, D^{t+1}, T^{t+1}, Y^{t+1}, B^{t+1}; g) \quad \# \quad (4)$$

Furthermore, to explore the sources of GLPI change, we decomposed the GLPI into efficiency change (EC) and technology change (TC), calculating this as

$$EC_n = \bar{D}_n^t(K^t, L^t, E^t, Z^t, D^t, T^t, Y^t, B^t; g) - \bar{D}_n^{t+1}(K^{t+1}, L^{t+1}, E^{t+1}, Z^{t+1}, D^{t+1}, T^{t+1}, Y^{t+1}, B^{t+1}; g) \quad (5)$$

$$TC_n = [\bar{D}_n^G(K^t, L^t, E^t, Z^t, D^t, T^t, Y^t, B^t; g) - \bar{D}_n^t(K^t, L^t, E^t, Z^t, D^t, T^t, Y^t, B^t; g)] - [\bar{D}_n^G(K^{t+1}, L^{t+1}, E^{t+1}, Z^{t+1}, D^{t+1}, T^{t+1}, Y^{t+1}, B^{t+1}; g) - \bar{D}_n^{t+1}(K^{t+1}, L^{t+1}, E^{t+1}, Z^{t+1}, D^{t+1}, T^{t+1}, Y^{t+1}, B^{t+1}; g)] \quad (6)$$

where $\bar{D}^t$ means using the technology frontier of t period when calculating the inefficiency value. The $\bar{D}^G$ means calculating the inefficiency

value by the global technology frontier. EC describes the change in the distance to the production frontier of province n during periods t and $t+1$. TC describes the change in the entire production frontier, which represents technology innovation.

Based on the additivity of the nonradial directional distance function and the Luenberger productivity index, and following Fujii et al. (2014), we can obtain the composition of GLPI.

Based on the results above, we obtain the contribution of each factor of production using the following formula,

$$\text{Contribution}_i = \frac{GLPI_i}{GLPI} \quad i = K, L, E, Z, D, T, Y, B\# \quad (8)$$

where $GLPI_i$ is the GLPI of i factor of production and $GLPI$ is calculated by equation (4).

3. Data

We collected panel data for 30 Chinese provinces (excluding Tibet, Hong Kong, Macao, and Taiwan) from 2010 to 2019 for analysis. We measure labor (L) by the amount of employment at the end of each year. Capital (K) is measured by the capital stock calculated by the perpetual inventory method. We assume that the rate of depreciation is 9.6% following Zhang et al. (2004), and we calculated the data by the real growth rate to replace the missing data. We measure energy inputs (E) using total energy consumption converted to standard coal. The data were collected from the China Energy Statistical Yearbook. We measure the desirable output (Y) by GDP, which was deflated based on the year of 2009, and collected the relevant data from the China Statistical Yearbook. Subsequently, the undesirable output (B) is measured by sulfur dioxide (S) and carbon emissions (C). The data for sulfur dioxide (SO_2) emissions is collected from the China Statistical Yearbook, and the data on provincial carbon emissions were collected from the Carbon Emission

Accounts & Datasets.

As for the intangible factors of production, data (D), knowledge (Z), and technology (T), considering the complex relationship among the intangible factors and the intangible and traditional factors, we obtain the intangible factors through factor analysis. Factor analysis is a commonly used and mature statistical method. It mainly focuses on reducing the dimension of a series of related variables to several unrelated common factors, which can well reflect the main information of all the original variables. The intangible factors of production used in our study are these common factors. Specifically, this study first collected 37 relevant variables at the provincial level in the range 2010–2019 from six dimensions: financial, research, education, publication, technology, and information; each of the six hierarchies is related to the intangible factors of production to some extent, and the details of these variables are shown in Table 1. Subsequently, we reduced the dimension of the index through factor analysis and named each part as three main factors:

Table 1

The indicators used in calculating the intangible factors of production.

Financial	Amount of media expenditure in financial expenditure
	Amount of education expenditure in financial expenditure
	Amount of science and technology expenditure in financial expenditure
Research	Employment in scientific research Average salary in scientific research field R&D funds of enterprises above designated size Number of R&D personnel of enterprises above designated size Number of projects of enterprises above designated size
Education	Number of secondary vocational teachers Number of secondary vocational school graduates Average wage of education industry Number of degrees awarded Employment in education industry Number of university teachers Number of colleges and universities
Publication	Total newspaper circulation Number of types of newly published books Number of types of published books Total number of periodicals
Technology	Patent authorization Number of patent applications Turnover of technology market Patent applications of enterprises above designated size
Information	Number of internet access households Number of internet ports Employment in information industry Average wage of information industry Optical cable length Number of domain names Number of digital TV subscribers Telecommunication business volume Telephone penetration rate Number of mobile telephone switches Number of web pages Mobile phone penetration rate Number of postal outlets Long-distance optical cable length

data, knowledge, and technology factors, according to the large factor loading of the variables. The data for the factor analysis were primarily collected from the China Statistical Yearbook, the National Bureau of Statistics, and provincial statistical yearbooks.

To unify the units of variables and avoid negative and zero values, we use deviation standardization for all variables above following Wang (1996) and Zhang and Xu (2022), and to avoid the influence of different lower limits, we set the lower limit to 0.0001, which is near to zero. The formula is

$$f_i = 0.0001 + 0.9999 \times \frac{x_i - \min x_i}{\max x_i - \min x_i} \quad f_i \in [0.0001, 1] \# \quad (9)$$

where f_i is the standardized variable; x_i is the score of factor analysis; and $\max x_i$ and $\min x_i$ are the maximum and minimum of x_i , respectively.

Table 2

The contribution rate of each input to TFP.

Year	Knowledge	Data	Technology	Capital	Labor	Energy
2011	−0.0243	0.0516	0.0509	−0.0091	0.0051	0.0316
2012	0.1500	−0.0051	0.2607	−0.0355	0.0838	0.0696
2013	0.1061	−0.0918	−0.0014	0.1221	−0.2090	−0.1980
2014	0.2503	0.0153	−0.0139	0.0573	−0.0496	−0.0132
2015	0.2780	0.0303	0.1636	−0.1112	0.0843	0.0604
2016	0.0416	0.0245	−0.0020	−0.0282	0.0253	0.0125
2017	−0.3024	−0.0474	−0.0347	−0.0355	0.0441	−0.0327
2018	−0.0052	0.0333	0.0232	−0.0280	0.0178	0.0383
2019	−0.1165	0.0177	0.0027	−0.0787	0.0412	−0.0374
Mean	0.0420	0.0032	0.0499	−0.0163	0.0048	−0.0077

4. Results

Based on the aforementioned equations, we obtain the results of the contribution of intangible factors of production to TFP and their temporal and spatial trends, along with the temporal variation trends and decomposition of TFP in China. The contribution of intangible factors of production is decomposed into EC and TC, while the contribution and change of the intangible factors of production to carbon total factor productivity (CTFP) are also explored.

4.1. The temporal and spatial trends of the contribution rate

Table 2 reports the results of the contribution rate of each production factor to TFP change in China. In general, the contribution of capital was essentially negative throughout the study period, with an average contribution rate of capital of −1.63%. This shows that China already

Table 3
Statistical checks.

	Simar-Wilson	OLS
Data	0.0437 (0.0288)	0.0544 (0.0333)
Technology	0.0691*** (0.0195)	0.0514** (0.0234)
Knowledge	0.0636*** (0.0231)	0.0516* (0.0295)
Capital	0.0049 (0.0121)	−0.0145 (0.0213)
Labor	0.0431 (0.0282)	0.0370 (0.0284)
Energy	0.0236 (0.0179)	0.0063 (0.0221)

Notes: Standard errors are reported in parentheses.

***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

has problems such as excessive investment (Bai and Xu, 2017); another reason is that China's TFP of capital presents a pro-cyclical feature; therefore, the negative TFP change of capital is consistent with the downward growth rate of China's economy in recent years, which also shows that China is no longer suitable for expanding investment to improve the quality of economic development. Meanwhile, the contribution level of traditional production factors, such as labor, is also low, an average of 0.48%, while the average contribution of energy was negative. Therefore, considering the weakness of traditional production factors in improving the quality of economic development, we must focus on improving the quality of China's economic development by using intangible production factors. In the following analysis, we focus on the contribution of the intangible factors of production.

For the intangible factors of production, the average contribution rates are positive. Knowledge, data, and technology annually contribute approximately 4.2%, 0.32%, and 5%, respectively, to the TFP during the study period. Based on the foregoing analysis, applying the intangible factors of production is an important way to improve high-quality economic development and outperform traditional factors of production.

Following Simar and Wilson (2007), this study simultaneously uses two-stage semi-parametric methods (Simar–Wilson) and ordinary least squares regression (OLS) to obtain corresponding statistical inference. First, we obtain the efficiency score without one of the factors through a nonparameter model; subsequently, we add the factor into the Simar–Wilson and OLS models and obtain the statistical inference. The results are shown in Table 3 and are consistent with the benchmark results on the whole. The effects of the intangible production factors (technology, knowledge, and data) in promoting high-quality economic development are statistically significant, whereas those of the traditional factors (labor, capital, and energy) are statistically insignificant. The results of statistical inference once again proved the robustness of the benchmark results.

Fig. 1 displays the TFP variation trend of each factor of production more intuitively and graphically. It shows that among the three factors of production, technology contributes the most to TFP growth followed by data and knowledge. Overall, the contribution of technology and knowledge show opposing change trends. We also notice that the cumulative growth of technology and data TFP are positive from 2011 to 2019, while that of knowledge is negative in certain years. The contribution of technology is still lower than that of technological change (Fig. 5), which indicates that the efficiency of technological innovation input in each province is lower than that of the country. Therefore, the central government can better promote high-quality economic development by pooling innovation resources at the national level. The cumulative trend of contribution to TFP of traditional production factors, such as capital, labor, and energy, are much lower than that of technology, while the negative impact of capital on high-quality economic development is increasing during the entire study period, which once again demonstrates the necessity and importance of applying intangible

factors of production.

China is vast, with great regional differences. We analyzed the heterogeneity from the perspective of geography, economic development level, and richness of relevant production factors. The 30 provinces can be divided by geographical location into southern (South) and northern (North),² the economic development level into higher (Developed) or lower (Developing) than average,³ and the richness of factors into higher (Sufficient) or lower (Deficient) than average.⁴ The results of the heterogeneity analysis are displayed in Fig. 2–4, where the horizontal axis is the year and the vertical axis is the cumulative TFP growth trend of each intangible factor of production.

Fig. 2 displays the cumulative knowledge TFP growth trend of different regions. The cumulative growth trend of knowledge exhibits the same trends in different regions in general but became different after 2017, and the cumulative knowledge TFP of most regions became negative, except for the developed and sufficient regions. Overall, compared with the southern, deficient and developing regions, the north, with sufficient and developed regions, performs better in the cumulative knowledge TFP. Moreover, the gap between the developing and developed regions is the largest, followed by sufficient and deficient regions and then northern and southern regions.

The cumulative data TFP growth trends in different regions are displayed in Fig. 3. The contribution of the data factor of production in the developed provinces is lower than that in the developing provinces, as well the sufficient and deficient regions. Except for the regions with sufficient data factor, the cumulative data TFP growth trends are positive in the last decade. Moreover, it is the sufficient regions that are demonstrating an increasing trend and becoming positive after 2017. In general, the cumulative growth trend of TFP in the data production factor shows great differences across different regions and growth was stable during the study period.

Fig. 4 displays the cumulative technology TFP growth trends across different regions. In general, the gap of cumulative technology TFP between different regions is small, especially after 2016, which means that the contribution of technology to economic high-quality development is not sensitive to geographic location, economic development level, and factor richness. The cumulative technology TFP in developed regions is higher than that of developing regions, which implies that the contribution of technology factor has a strong relationship with the economic development level. Initially, the technology factor in the northern regions performed better than the southern regions but the southern regions surpassed the north after 2014. Moreover, the cumulative contribution of technology positively performs at a high level, which announced again that technological innovation is the largest driver of economic high-quality development.

4.2. TC and EC analysis

Fig. 5 shows that the change in TFP (in orange) is positive in most

² The north provinces include Heilongjiang, Jilin, Inner Mongolia, Liaoning, Beijing, Tianjin, Hebei, Shandong, Henan, Shanxi, Shaanxi, Ningxia, Xinjiang, and Qinghai, while the other provinces are in the south.

³ The developed provinces are Shandong, Jiangsu, Shanghai, Guangdong, Henan, Anhui, Hunan, Hubei, Sichuan, Hebei, Beijing, Fujian, and Zhejiang, with the remaining provinces defined as developing provinces.

⁴ Provinces sufficient in the knowledge factor of production are Shanghai, Beijing, Shandong, Jiangsu, Guangdong, Henan, Anhui, Hunan, Hubei, Sichuan, Hebei, Jilin, Liaoning, Shaanxi, and Jiangxi, with the remaining provinces defined as deficient in the knowledge factor. Provinces with sufficient data factors of production include Beijing, Shanghai, Guangdong, Jiangsu, Tianjin, Zhejiang, Fujian, and Hainan, while the remainder represents provinces deficient in the data factor. Provinces sufficient in the technology factor of production include Shandong, Jiangsu, Shanghai, Zhejiang, Fujian, Guangdong, Henan, Anhui, Hunan, Hubei, Sichuan, and Hebei, while the remainder are provinces deficient in the technology factor.

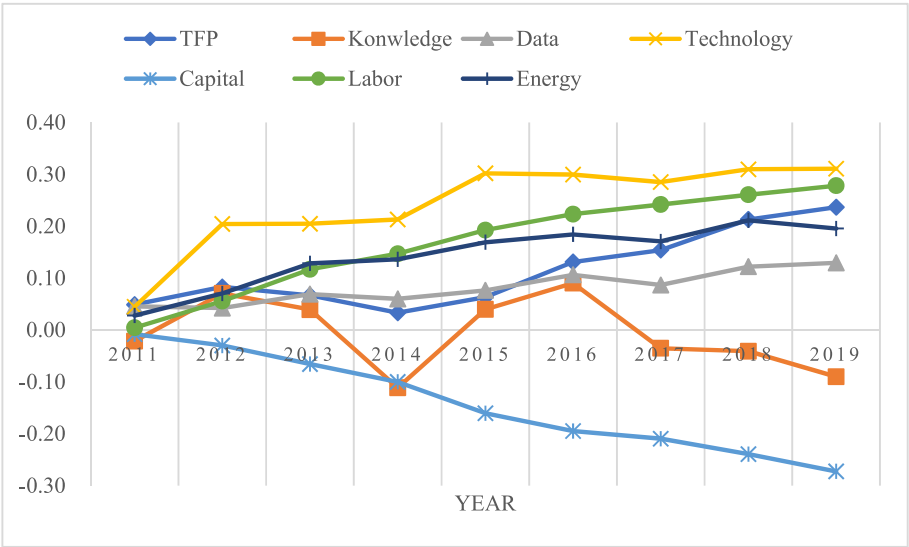


Fig. 1. Cumulative TFP variation trend of each input.

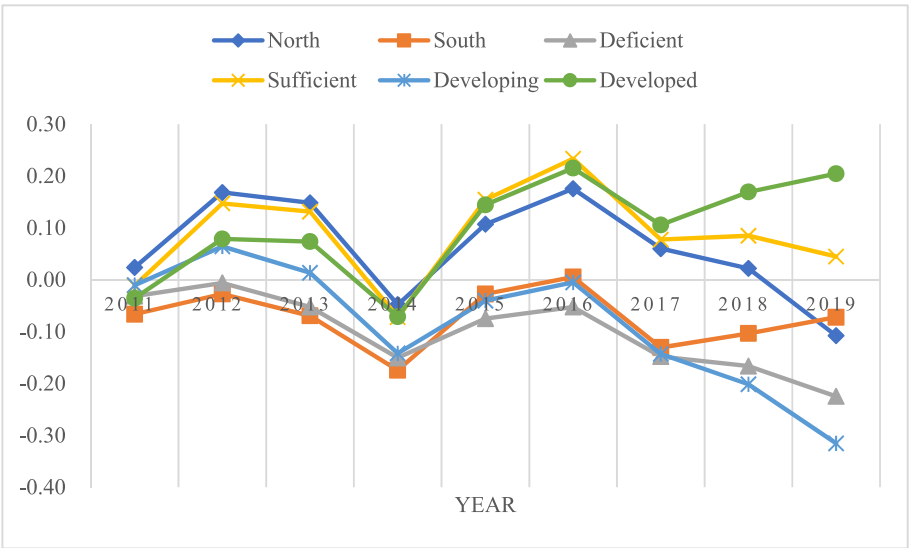


Fig. 2. Cumulative knowledge TFP growth trend in different regions.

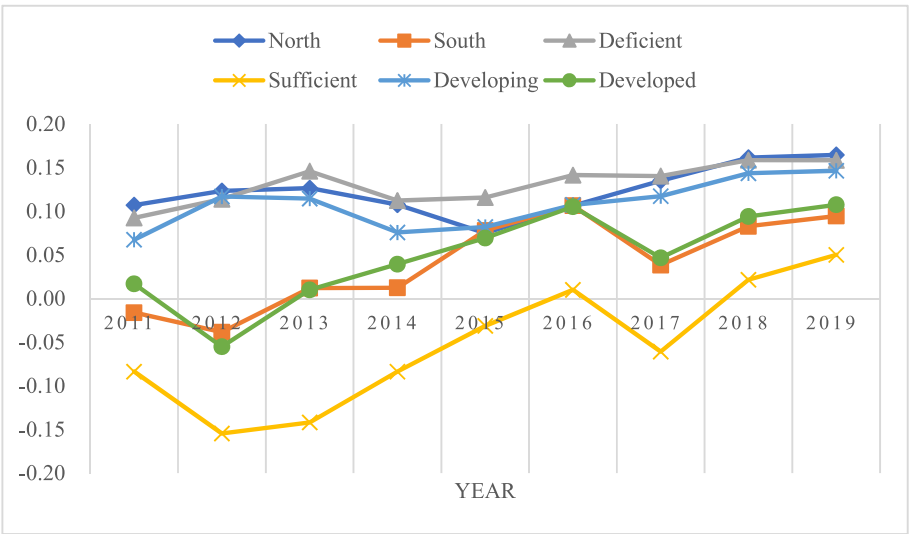


Fig. 3. Cumulative data TFP growth trends in different regions.

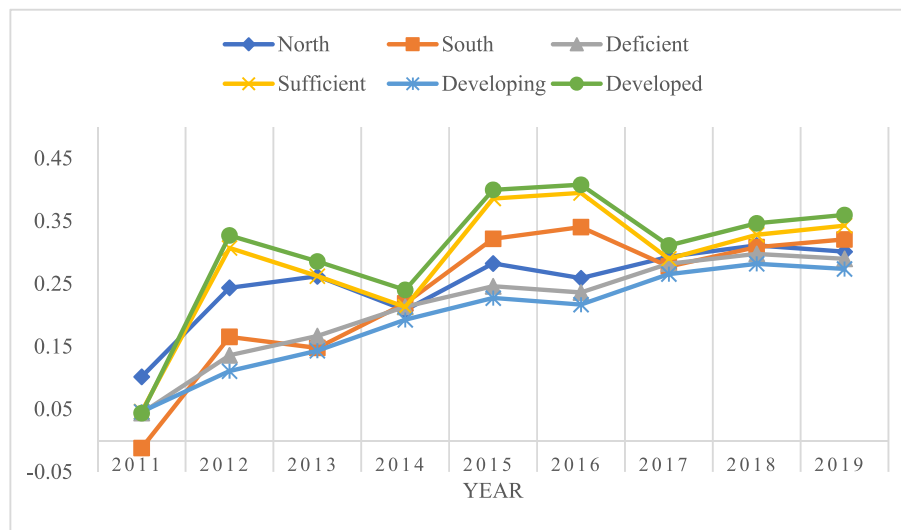


Fig. 4. Cumulative technology TFP growth trends in different regions.

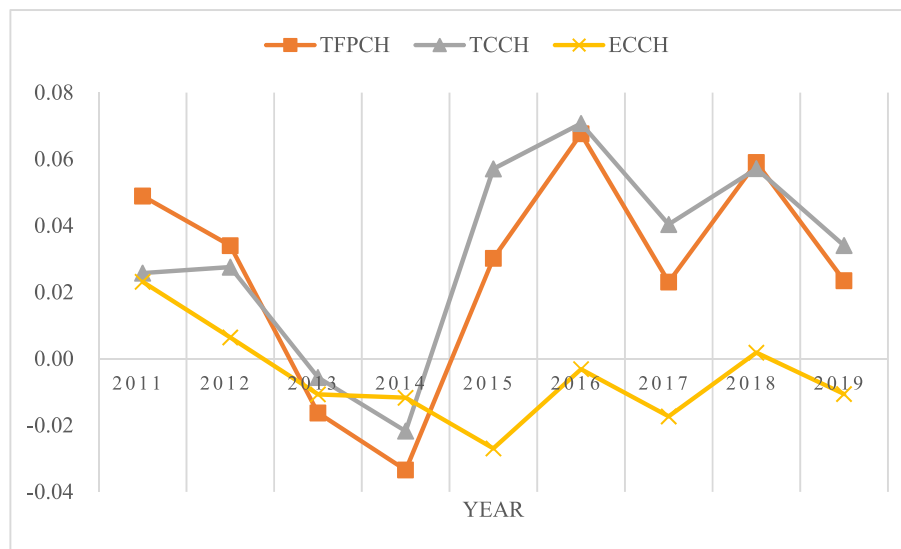


Fig. 5. Temporal variation trend and decomposition of TFP in China.

Table 4
The contribution rate of input to TC.

Year	Knowledge	Data	Technology	Capital	Labor	Energy
2011	-0.0991	-0.1404	-0.0053	-0.0850	-0.0309	-0.0522
2012	-0.0166	-0.0445	0.2488	-0.0758	0.0860	0.0746
2013	1.1882	0.6224	-1.4769	0.1396	-0.5384	-0.4768
2014	-0.0398	0.0541	0.2723	0.0638	-0.0487	-0.0140
2015	0.0283	0.0530	0.1464	-0.0258	0.0719	0.0376
2016	0.0151	0.0085	0.0263	-0.0170	0.0141	0.0173
2017	0.0898	-0.0319	-0.1618	-0.0231	0.0183	-0.0506
2018	0.0114	0.0269	0.0592	-0.0292	0.0114	0.0419
2019	0.1798	-0.0342	0.0225	-0.0197	-0.0015	0.0067
Mean	0.1508	0.0571	-0.0965	-0.0080	-0.0464	-0.0462

years, meaning that the quality of China's economic development is constantly improving. The overall change in TFP is relatively stable but a steep rise occurred in 2015. The main reason is that the Chinese government issued a Guideline on Measures to Boost Mass Entrepreneurship and Innovation in June 2015, which greatly improved the overall innovation level of China and promoted the TFP through improved

technical efficiency. For the results of TC (in gray) and EC (in yellow), the overall change in technical efficiency is positive and greatly fluctuates with time, while the overall change in efficiency is negative and the fluctuation is small and below 0. The change in technical efficiency directly determines the size and direction of the change in TFP, which is consistent with [Liu et al. \(2020\)](#) and [Li and Liu \(2015\)](#). Based on the

Table 5
The contribution rate of input to EC.

Year	Knowledge	Data	Technology	Capital	Labor	Energy
2011	0.0590	0.2653	0.1133	0.0753	0.0452	0.1249
2012	0.8629	0.1633	0.3116	0.1369	0.0745	0.0484
2013	-0.4629	-0.4674	0.7746	0.1129	-0.0357	-0.0515
2014	0.7899	-0.0568	-0.5460	0.0451	-0.0512	-0.0116
2015	-0.2516	0.0784	0.1272	0.0700	0.0581	0.0120
2016	-0.5632	-0.3420	0.6419	0.2264	-0.2309	0.1236
2017	0.6124	-0.0113	-0.3311	-0.0065	-0.0160	-0.0744
2018	-0.5134	0.2289	-1.0786	0.0077	0.2151	-0.0724
2019	0.8379	-0.1494	0.0665	0.1113	-0.0963	0.1047
Mean	0.1523	-0.0323	0.0088	0.0866	-0.0042	0.0226

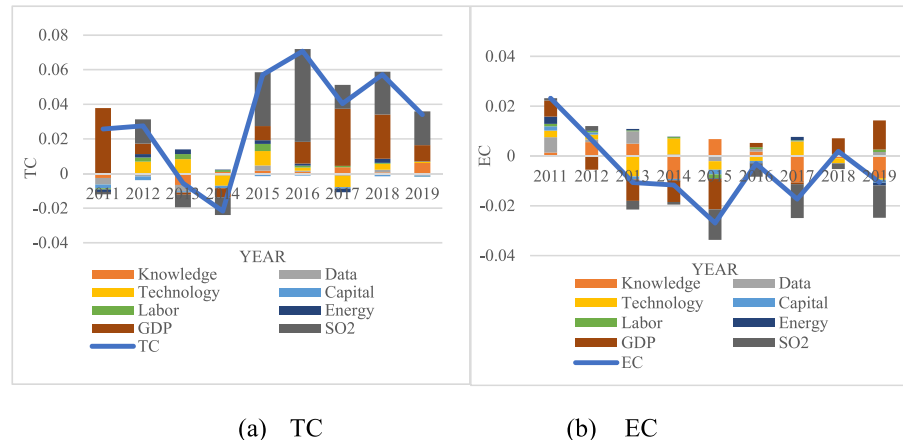


Fig. 6. Temporal variation trend and decomposition of TC and EC.

Table 6
The contribution rate of input to CTFP.

Year	CTFP	Knowledge	Data	Technology	Capital	Labor	Energy
2011	0.0482	-0.0070	0.0281	0.0417	-0.0270	0.0072	0.0358
2012	0.0397	0.1765	0.0516	0.1839	-0.0421	0.0559	0.0596
2013	0.0003	-2.7640	0.6629	1.0022	-7.9281	10.6718	10.7363
2014	-0.0147	0.6587	0.0676	0.0268	0.1915	-0.0217	0.0468
2015	0.0310	0.2527	0.0152	0.1405	-0.0955	0.0928	0.0651
2016	0.0293	0.0992	0.0282	0.0189	-0.0362	0.1052	0.0614
2017	0.0225	-0.0265	-0.0210	0.0101	-0.0130	-0.0148	0.0369
2018	0.0511	-0.0314	0.0281	0.0430	-0.0073	0.0405	0.0415
2019	0.0179	0.0124	0.0131	0.0858	0.0297	0.1252	0.0523
Mean	0.0250	-0.1810	0.0971	0.1725	-0.8809	1.2291	1.2373

aforementioned results, technological innovation remains the most important driver of China's high-quality development. Measures such as improving institutions should be taken to improve the level of EC to improve the TFP.

Table 4 shows that technical efficiency is always positive within the research interval, indicating that technical efficiency promotes TFP. The contribution to the average technical efficiency changes of intangible factors such as knowledge, data, and technology is 15.1%, 5.7%, and -9.7%, respectively. In general, the TC of intangible production factors is greater than that of traditional production factors.

Table 5 displays the contribution rate of factors of production to EC. The EC is alternating between positive and negative. The contribution of intangible factors of production such as knowledge, data, and technology to the change in the average efficiency is 15.2%, -3.2%, and 0.88%, respectively. During the study period, the average EC of labor is negative, whereas the EC of capital and energy are positive.

In general, the TC and EC levels of each intangible factor were similar and positive, but the contribution rate of labor and energy factors of production are insignificant, implying that the intangible factors of

production are the main contributors to EC and enhance the TFP through technical efficiency and increasing efficiency levels. Combining the results of Tables 4 and 5, the TC and EC of the knowledge and technology factors of production generally show a reverse change trend and alternate between positive and negative changes.

Fig. 6 presents the results of the temporal variation trends and decomposition of TC and EC. The TC is nearly positive, while the EC is nearly negative during the last decade. In Fig. 6, panel (a), the TC of SO₂ emissions is positive and large in scale and may be considered as the main driver of TC increases, which means that China has made great progress in technological innovation in pollutant reduction in recent years. The main driver of EC growth is the intangible factors of production, such as knowledge and technology (Fig. 6, panel (b)).

4.3. The contribution of input to CTFP

The aforementioned analysis uses SO₂ emissions as the undesirable output. In September 2020, Chinese President Xi Jinping announced at the United Nations that China would achieve its carbon emissions peak

by 2030 and be carbon neutral by 2060, termed the dual carbon goal. This is a concrete manifestation of China's economic transition to high-quality development, and will have a key impact on China's production and lifestyles. As the largest source of carbon emissions, China's dual carbon goal will also significantly affect the world. Considering the impact of China's dual carbon goal, we replace SO₂ emissions with carbon emissions generated in production and life as the undesirable output to study the impact of each intangible factor of production on CTFP (Table 6).

Table 6 shows that the CTFP is positive from 2011 to 2019, which implies constant improvement in high-quality development from the perspective of carbon emissions. The results of treating carbon emissions as the undesired output are consistent with the benchmark results for the contribution of each intangible factor to TFP. The contribution rate of changes in the intangible factors of production of knowledge, data, and technology is -18.1%, 9.7%, and 17.3%, respectively, while the contribution rate of capital remains negative. These results imply that the application of intangible factors improves the quality of economic development and accelerates the carbon reduction and realization of the dual carbon goal.

5. Conclusions

Based on the scientific measurement of each intangible factor of production, this study measures the drivers of China's high-quality economic development by incorporating each intangible factor of production into a nonparametric model, and we draw three main conclusions. First, the quality of China's economic development has been improving during the study period, where technological innovation is the largest driver, and the innovation efficiency at the national level is better than at the provincial level. Second, from the perspective of production factors, the performance of intangible factors of production is clearly better than that of traditional factors in promoting economic quality and realizing the dual carbon goal. Intangible factors of production such as knowledge, data, and technology have a promoting effect on TFP of about 4.2%, 0.32%, and 5%, respectively. In contrast, the capital factor of production has a negative impact on high-quality economic development. Third, the variation in the contribution of all factors of production to high-quality economic development differs by geographical location, economic development level, and production factor richness.

Our conclusions suggest several policy implications. First, China should continue to promote scientific and technological innovation to achieve high-quality economic development. The central government should coordinate various innovation resources and concentrate on mastering key technologies to improve the quality of economic development. Second, in terms of factors of production, China should actively promote the wide application of intangible factors of production and improve traditional factors, such as labor and capital, to realize the transition to high-quality economic development and the nation's dual carbon goal. In particular, China should optimize its investment structure to avoid the negative impact of capital on high-quality economic development. Third, owing to differences in resource endowments and level of economic development in different regions, the structure of various factors of production should be allocated based on local realistic conditions.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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